IOC Olympic Data Platform

Glintt Next Recruitment Process - Study / Assessment Case

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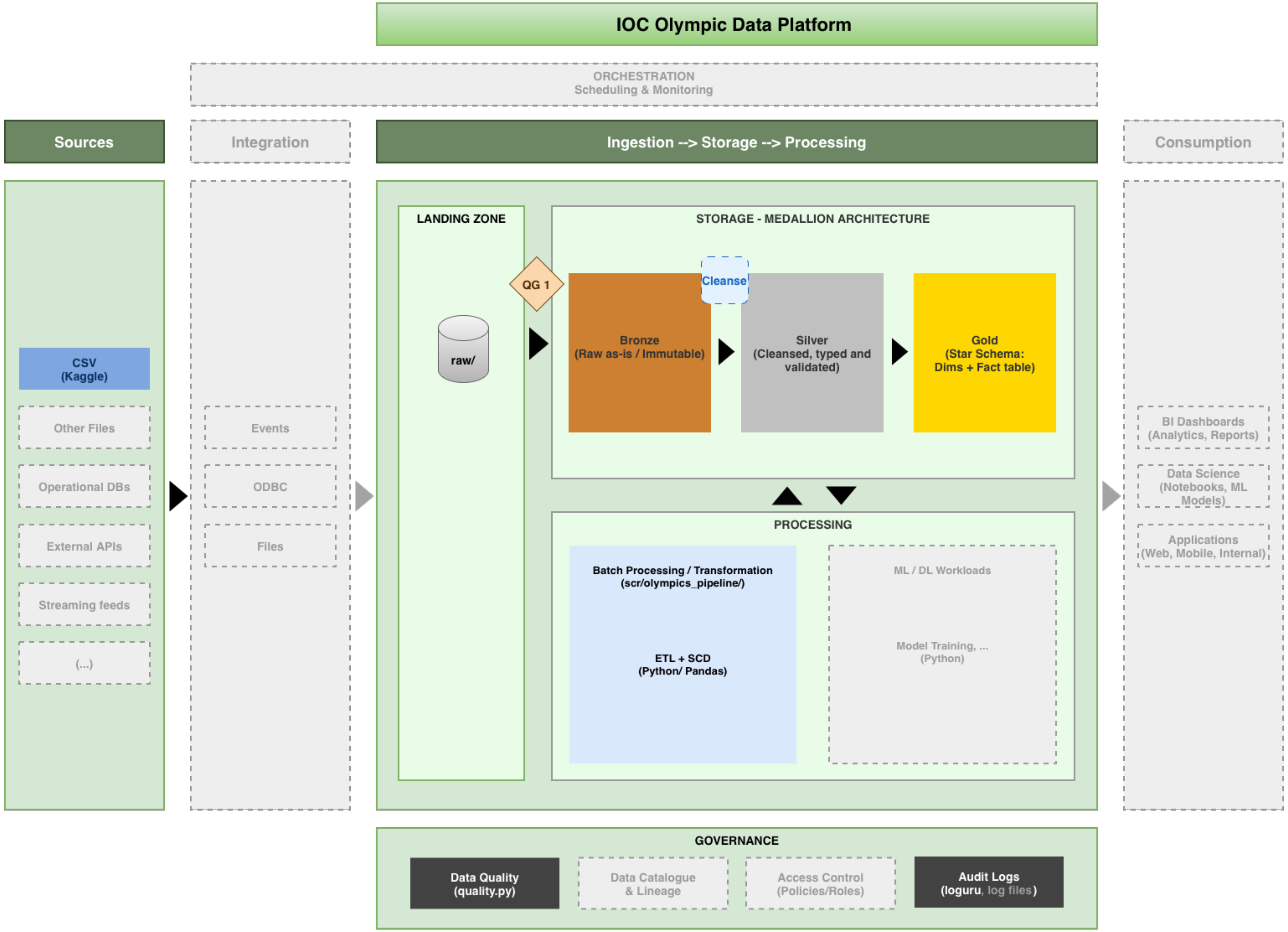
Data Platform Architecture



Out of scope for this assessment. Architectural reference only to illustrate an example of production architecture.

Quality Gate 1: quality.py (Drops structurally invalid rows before Bronze ingestion)

Cleansing Step: build_silver() (Cleanses, types and standardises data for the Silver layer)

Architecture Decision — Executive Summary

The proposed platform **adopts a *medallion* data lake architecture** built entirely on open-source tooling and open formats.

Data progresses through three layers:

- ***Bronze*** for raw, immutable ingestion;
- ***Silver*** for cleansed and typed data;
- ***Gold*** for the analytical *star schema.*

Together, they ensure full auditability and a clear separation of concerns across all data layers.

A **quality gate rejects structurally invalid records** before *Bronze* ingestion and a subsequent **cleansing step standardizes and types data** at the *Bronze-to-Silver* transition.

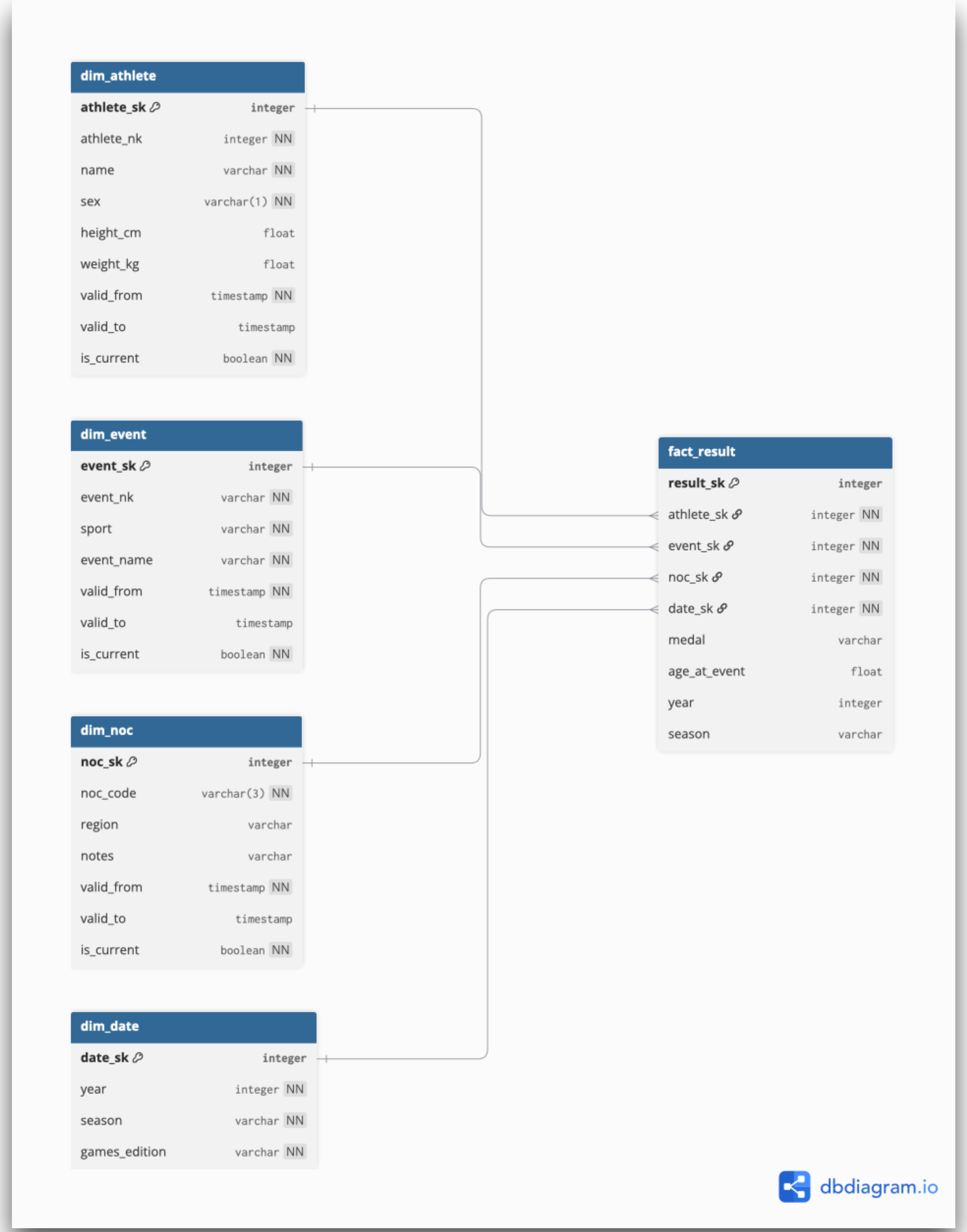
SCD Type 2 **is applied at the *Silver-to-Gold* transition**, versioning dimension records over time and preserving the full historical profile of athletes, events, and countries across 120 years of Olympic history.

The architecture supports analytics, machine learning, and deep learning workloads through a shared data lake, eliminating data duplication across teams and enabling consistent feature access.

The governance layer implemented covers data quality gates and structured audit logging. The architecture is designed to extend to lineage, cataloguing, and access control in production.

This assessment delivers a fully working local implementation: *pandas* replaces *PySpark*, and local *Parquet* files replace a distributed object store, but the *Bronze-Silver-Gold* separation, *SCD Type 2*, quality validation, typed code (*mypy*), and unit tests are production-grade. The quality gates, validations, and tests presented here are intentionally scoped to the assessment dataset and representative scenarios, each is designed to scale horizontally in a production environment with additional rules and coverage as the platform grows.

Star Schema — Olympic Data Warehouse



SCD 0: Immutable | SCD 1: Overwrite | SCD 2: History tracked

Slowly Changing Dimensions - Column Classification

Table	Column	SCD Type	Rationable
DIM_ATHLETE	*name*	Type 1	Overwrite, name corrections (Not historical events)
DIM_ATHLETE	*sex*	Type 1	Overwrite (No historical value)
DIM_ATHLETE	*height_cm*	Type 2	Physical changes matter for analysis
DIM_ATHLETE	*weight_kg*	Type 2	Weight changes affect performance analysis
DIM_EVENT	*sport*	Type 2	Sports reorganised across Olympic editions
DIM_EVENT	*event_name*	Type 2	Event names changed over 120 years
DIM_NOC	*region*	Type 2	Countries renamed, unified or dissolved
DIM_NOC	*notes*	Type 1	Administrative (No historical value)
DIM_DATE	*all columns*	Type 0	Olympic dates are immutable

Implemented: SCD Type 2 full pipeline in scd.py